

5.3 DIAGONALIZATION

$$D_2 = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$D_3 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I = \text{diag.} \quad \bar{O} =$$

Def: A matrix is called diagonal matrix if

$$d_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ d & \text{if } i = j \end{cases}, \text{ where } d \in \mathbb{R}$$

$$D^2 = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 2^2 & 0 \\ 0 & 3^2 \end{bmatrix}$$

$$D^3 = \begin{bmatrix} 2^2 & 0 \\ 0 & 3^2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

$$D^k = \begin{bmatrix} d_{11}^k & & 0 \\ & d_{22}^k & \\ 0 & & \ddots \\ & & & d_{nn}^k \end{bmatrix}$$

Property #1

(Power of diag. mat)

$$|D - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & 3-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(3-\lambda) = 0 \Rightarrow$$

$$\begin{cases} \lambda_1 = 2 \\ \lambda_2 = 3 \end{cases}$$

Property #2

E/values

E/values of a diag. matrix are its entries.

1 A^n (Fast calculation)

Let's suppose that $\exists$ invertible P :

$$A = P \cdot D \cdot P^{-1}$$

Then $A^n = \underbrace{P \cdot D \cdot P^{-1}}_{2 < n}$

D. $A^n = \underbrace{A \cdot A \cdot A \cdot \dots \cdot A}_n =$

$$= (P \cdot D \cdot P^{-1}) \cdot (P \cdot D \cdot P^{-1}) \cdot \dots \cdot (P \cdot D \cdot P^{-1}) =$$

$$= P \cdot \underbrace{D \cdot D \cdot D \cdot \dots \cdot D}_n \cdot P^{-1} = P \cdot D^n \cdot P^{-1}$$

$$\underbrace{A \cdot A \cdot A \cdot \dots \cdot A}_{n \text{ times}} \Rightarrow$$

$$A \cdot A = O(n^3)^{\text{size}}$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

$$P^{-1} \cdot P = I$$

Def: Two matrices are called similar if $\exists$ invertible matrix P :

A, B

$$A = P \cdot B \cdot P^{-1}$$

If two matrices are similar then their e/values are the same

D. $|A - \lambda \cdot I| = |P \cdot B \cdot P^{-1} - \lambda \cdot P \cdot P^{-1}| = |P \cdot B \cdot P^{-1} - P \cdot \lambda \cdot P^{-1}| =$

$$= |P \cdot (B \cdot P^{-1} - \lambda \cdot P^{-1})| = |P \cdot (B - \lambda \cdot I) \cdot P^{-1}| =$$

$$= |P| \cdot |B - \lambda I| \cdot |P^{-1}| = |B - \lambda I|$$

$$|A \cdot B| = |A| \cdot |B|$$

$$|P| \cdot |P^{-1}| = |I| = 1$$

$\Rightarrow A$ and B have the same e/values.

So If B is diag. $\Rightarrow$ e/values of A are found, they are the entries of B

EXAMPLE 2 Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a formula for A^k , given that $A = PDP^{-1}$, where

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \text{ and } D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

$$A^k = P \cdot D^k \cdot P^{-1}$$

$$D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix},$$

$$|P| = \begin{vmatrix} 1 & 1 \\ -1 & -2 \end{vmatrix} = -2 + 1 = \textcircled{-1}$$

$$P^{-1} = \frac{1}{|P|} \cdot \begin{bmatrix} p_{11} & p_{21} \\ p_{12} & p_{22} \end{bmatrix},$$

$$p_{12} = (-1)^{1+2} \cdot (-1) = 1$$

$$p_{21} = (-1)^{2+1} \cdot 1 = -1$$

$$p_{11} = (-1)^{1+1} \cdot (-2) = -2;$$

$$p_{22} = (-1)^{2+2} \cdot 1 = 1$$

$$P^{-1} = \frac{1}{-1} \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

$$P \cdot P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^k = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an **eigenvector basis** of $\mathbb{R}^n$.

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$$P = \begin{pmatrix} \bar{v}_1 & \bar{v}_2 & \dots & \bar{v}_n \end{pmatrix}$$

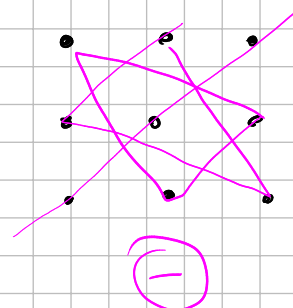
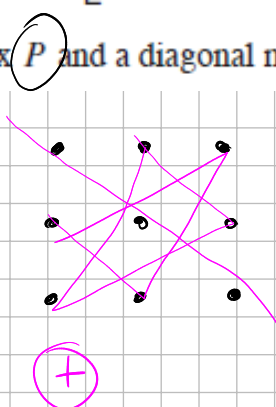
Diagonalizing Matrices

EXAMPLE 3 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

$$\begin{vmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{vmatrix} = 0$$



$$(1-\lambda)(-5-\lambda)(1-\lambda) + 3 \cdot 3 \cdot (-3) + (-3) \cdot 3 \cdot 3 - 3 \cdot (-5-\lambda) \cdot 3 - 3 \cdot (-3) \cdot (1-\lambda) - 3 \cdot (-3) \cdot (1-\lambda) = 0$$

$$(1-2\lambda+\lambda^2)(-5-\lambda) + 54 - 9(-5-\lambda) + 18(1-\lambda) = 0$$

$$-\lambda^3 - 3\lambda^2 + 9\lambda - 5 - 54 + 45 + 9\lambda + 18 - 18\lambda = 0$$

$$-\lambda^3 - 3\lambda^2 + 4 = 0$$

$$\lambda_1 = 1$$

$$-1^3 - 3 \cdot 1^2 + 4 = -1 - 3 + 4 = 0 \quad \checkmark$$

$$(\lambda-1)(\lambda^2 + 4\lambda + 4) = 0$$

$$(\lambda-1)(\lambda+2)^2 = 0 \quad \lambda_1 = 1, \lambda_2 = -2$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$\lambda_1 = 1 \quad \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} - 1 \cdot I = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \Rightarrow$$

$$\begin{array}{r|l} \lambda^3 + 3\lambda^2 - 4 & \lambda - 1 \\ \lambda^3 - \lambda^2 & \lambda^2 + 4\lambda + 4 \\ \hline 4\lambda^2 - 4 & \\ -4\lambda^2 - 4\lambda & \\ \hline 4\lambda - 4 & \\ -4\lambda - 4 & \\ \hline 0 & \end{array}$$

$$\lambda_1 = 1, \quad \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_3 \quad R_2 + R_3 \quad R_3 / 3$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \quad \begin{array}{l} R_1 - R_2 \\ R_2 / -3 \\ R_3 + R_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{cases} x_1 - x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \quad \begin{array}{l} \uparrow \text{free} \\ x_3 = 1 \end{array}$$

$$\bar{x} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$P^{-1} \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} R_2 + R_1 \\ R_3 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{bmatrix} \quad R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & 1 & 0 \end{bmatrix} \quad R_1 + R_2$$

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & 1 & 0 \end{pmatrix} \quad \begin{array}{l} R_1 + R_3 \\ R_2 + 2R_3 \\ R_3 / -1 \end{array}$$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{pmatrix} \quad \begin{array}{l} \text{I} \\ P^{-1} \end{array}$$

$$A = P D P^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{pmatrix} \quad \leftarrow P^{-1}$$

$$\lambda_2 = -2$$

$$\begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 0$$

basic free

$$x_2 = s \quad x_3 = t \Rightarrow$$

$$\bar{X} = \begin{bmatrix} -x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -s - t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

2 e/vectors

lin indep vectors

EXAMPLE 4 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

SOLUTION The characteristic equation of A turns out to be exactly the same as that in Example 3:

$$0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$. However, it is easy to verify that each eigenspace is only one-dimensional:

Basis for $\lambda = 1$: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Basis for $\lambda = -2$: $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

There are no other eigenvalues, and every eigenvector of A is a multiple of either $\mathbf{v}_1$ or $\mathbf{v}_2$. Hence it is impossible to construct a basis of $\mathbb{R}^3$ using eigenvectors of A . By Theorem 5, A is not diagonalizable. ■

The following theorem provides a *sufficient* condition for a matrix to be diagonalizable.

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.